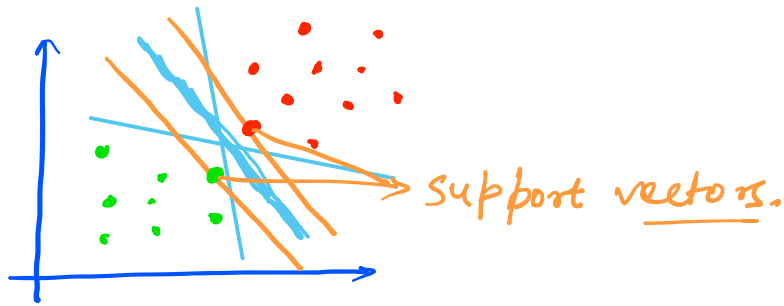


Support Vector Machine

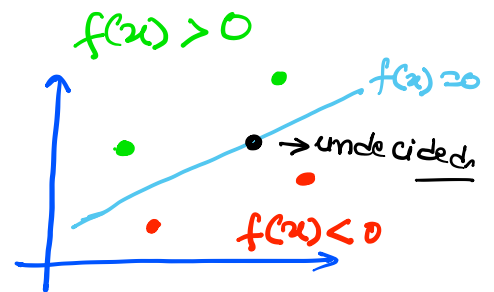


① High dimensional well defined

② Deals with outliers

Assume, $y \in \{-1, 1\}$

$$w_1x_1 + w_2x_2 + \dots + b$$
$$w^T x + b$$



In 2D,

$$f(x) = w_1x_1 + b = 0 \text{ (boundary equation)}$$

$$f(x) = w^T x + b$$

$$\hat{y} = \text{sign}(f(x))$$

score	prediction	meaning.
$f(x) > 0$	positive (+1)	points on the +ve side
$f(x) < 0$	negative (-1)	points on the -ve side
$f(x) = 0$	on the boundary	undecided.

Example:

$$f(x) = w^T x + b = +4.31 = \text{sign}(4.31) = +1$$

$$f(x) = w^T x + b = -10.9 = \text{sign}(-10.9) = -1$$

Margin Lines:

$$y \in \{+1, -1\}$$

correct classification

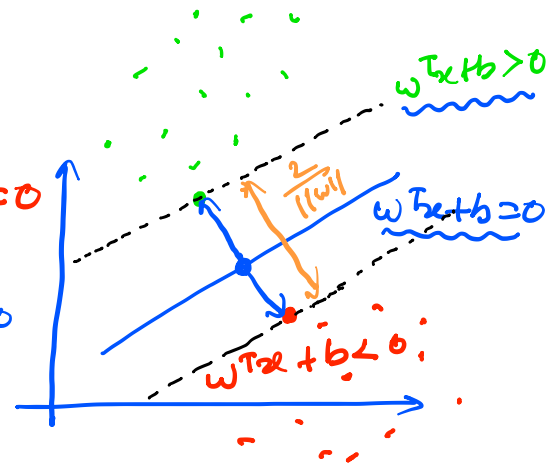
$$y_i (w^T x_i + b) = 1$$

y_i	$w^T x + b$	
+1	+1	1 ✓
-1	-1	1 ✓
+1	-1	-1 ✗
-1	+1	-1 ✗

How far is $w^T x + b > 0$ from $w^T x + b = 0$

Distance from point x to $w^T x + b = 0$

$$\frac{|w^T x + b|}{\|w\|}$$



for the positive margin line $\text{sign}(w^T x + b) = 1$

$$\text{distance} = \frac{1}{\|w\|}$$

for the -ve margin also,

$$\text{distance} = \frac{1}{\|w\|}$$

$$\text{margin width} = \frac{1}{\|w\|} + \frac{1}{\|w\|} = \frac{2}{\|w\|}$$

multiple decision boundaries will give multiple margin width. The line which maximizes the margin, is the line we are looking for.

① Hard margin objective:

$$\text{maximize } \frac{2}{\|w\|} \quad \downarrow \quad \uparrow$$

$$\text{minimize } \|w\|$$

$$\boxed{\min_{w, b} \frac{1}{2} \|w\|^2}$$

$$\text{s.t. } y_i (w^T x_i + b) \geq 1$$

* Hinge Loss:

$$L_i = \max(0, (1 - y_i (w^T x + b)))$$

Loss	y_i	$w^T x + b$
0	+1	+10.1
11.1	+1	-10.1

$$\begin{aligned} &\rightarrow 1 - +1(10.1) \\ &1 - 10.1 = -9.1 \\ &\max(0, -9.1) \end{aligned}$$

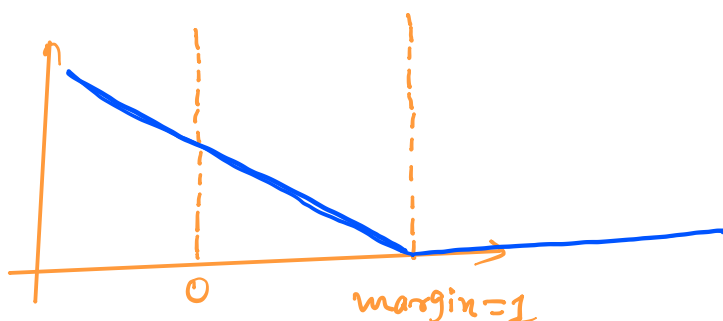
$$= 0$$

$$\begin{aligned} &1 - +1(-10.1) \\ &1 - (-10.1) \end{aligned}$$

$$1 + 10.1 = 11.1$$

$$\max(0, 11.1) = 11.1$$

$$\text{Hinge loss} = \max(0, 1 - \text{margin})$$



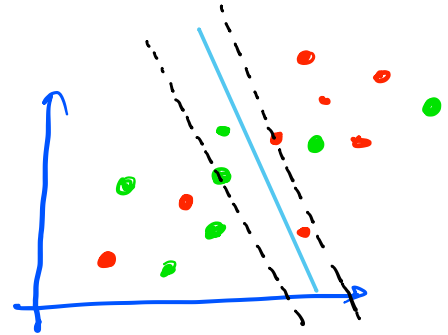
$$J(w, b) = \underbrace{\frac{1}{2} \|w\|^2}_{\substack{\downarrow \\ \text{prefers a} \\ \text{wide margin}}} + C \cdot \underbrace{\frac{1}{m} \sum_{i=1}^m \max(0, 1 - y_i(w^T x_i + b))}_{\substack{\text{controls how expensive the} \\ \text{violations are,} \\ \text{violations,}}}$$

Gradient descent update:

$$w := w - \eta \cdot \frac{\partial J}{\partial w}, \quad b := b - \eta \cdot \frac{\partial J}{\partial b}$$

* Soft-margin SVM

* Slack variable,



$$\underbrace{y_i(w^T x_i + b)}_{\substack{\text{Slack variable,} \\ \xi_i \geq 0}} \geq 1 - \xi_i$$

Slack value	meaning.
$\xi_i = 0$	point is correctly classified & outside the margin
$0 < \xi_i < 1$	point is inside boundary but still on the <u>correct side</u> ,
$\xi_i \geq 1$	point is <u>misclassified</u> ,

Soft margin objective:

$$\min \frac{1}{2} \|w\|^2 + C \sum_i \epsilon_i$$

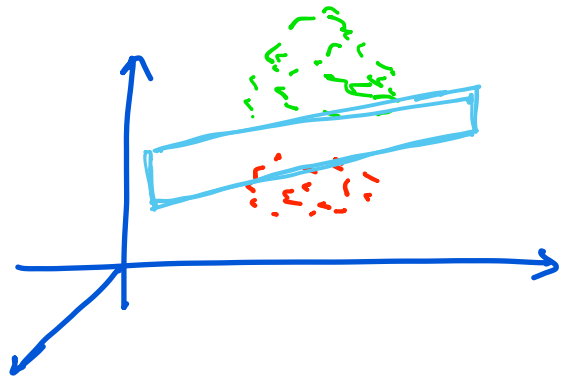
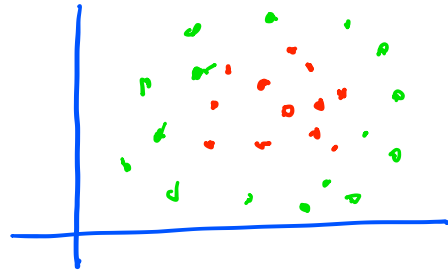
C	behaviour	
small	allowing more <u>penalties</u>	underfit
large	stricter, trying to reduce the violations.	<u>overfit</u> .

* Kernel-trick

Feature mapping:

$$x \rightarrow \phi(x)$$

$$w^T \phi(x) + b = 0$$



- ① Polynomial kernel
defined by a degree

$$x_1^2, x_2^2, x_1 \cdot x_2$$

$$K(x_i, x_j) = (x_i^T x_j + \gamma)^d$$

- ② RBF kernel

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2)$$

when to use SVM:

- ① small to medium feature dataset
- ② High dimensional $\rightarrow$ Linear SVM

when not to use SVM:

- ① need prob. estimate
- ② millions of rows in the dataset
- ③ explainability is hard

mandatory:

Feature scaling,

for multiclass SVM

- ① one vs. rest
- ② one vs. one